A Machine Learning Approach to Economic Growth Prediction Using XGBoost

Abstract

This paper presents a machine learning-based approach to predict economic growth rates across multiple countries using historical economic indicators such as GDP, inflation rate, and unemployment rate. Employing the XGBoost algorithm, the study develops a predictive model, evaluates its performance, and analyzes feature importance using SHAP values. The results demonstrate the potential of machine learning to enhance the accuracy of economic forecasting, offering insights for policymakers and researchers. Challenges such as data quality and external economic factors are addressed, and future directions for improving prediction models are proposed.

1 Problem Statement

Accurate prediction of economic growth is essential for effective policymaking, investment decisions, and economic planning. Traditional forecasting methods, such as econometric models, often struggle with the volatility of economic indicators, limited historical data, and the influence of external factors like geopolitical events. This study aims to address these challenges by developing a machine learning model to predict economic growth rates, leveraging historical data and advanced algorithms to improve forecasting accuracy.

2 Introduction

2.1 Essential Background of Economic Indicators

Economic growth, typically measured as the annual percentage change in Gross Domestic Product (GDP), reflects a country's economic health. Key indicators influencing growth include:

- GDP: The total monetary value of goods and services produced within a country.
- Inflation Rate: The rate at which the general level of prices for goods and servicesincreases.
- Unemployment Rate: The percentage of the labor force that is unemployed andactively seeking work.

These indicators provide critical insights into economic trends and are used in forecasting models.

2.2 How Economic Data is Collected and Used

Economic data is collected by organizations such as the World Bank, International Monetary Fund (IMF), and national statistical agencies. This data is used to analyze economic performance, inform policy decisions, and develop forecasting models. In this study, a dataset spanning 2010 to 2025 across 19 countries is used, including variables like GDP, inflation, and unemployment.

2.3 Importance of Economic Growth Prediction

Predicting economic growth is vital for governments to design fiscal and monetary policies, for businesses to plan investments, and for individuals to make informed financial decisions. Accurate forecasts can mitigate risks associated with economic volatility and support sustainable development.

2.4 Motivation of Work

Traditional forecasting methods, such as ARIMA and econometric models, often fail to capture complex, non-linear relationships in economic data. Machine learning, particularly ensemble methods like XGBoost, offers a promising alternative by handling large datasets and identifying intricate patterns. This project is motivated by the need to improve forecasting accuracy and provide interpretable insights into economic growth drivers.

2.5 Scope of the Study

This study focuses on predicting economic growth rates using a dataset of economic indicators from 19 countries (2010–2025). The XGBoost algorithm is employed, with features including lagged values of GDP, inflation, and unemployment. The study evaluates model performance and uses SHAP for feature importance analysis.

3 Literature Review

Economic forecasting has traditionally relied on econometric models like ARIMA and vector autoregression (1). However, these methods struggle with non-linear relationships and external shocks. Recent studies have explored machine learning for economic forecasting:

- (author?) (2) used mixed-frequency vector autoregressive modeling to forecast China's GDP growth, comparing results with expert predictions.
- A 2024 study on international trade networks utilized machine learning to enhanceGDP growth predictions by incorporating topological measures (3).
- Ensemble methods and deep learning models have shown improved accuracy ineconomic forecasting, with tools like SHAP enhancing interpretability (4).

This study builds on these advancements by applying XGBoost to predict economic growth and analyzing feature importance.

4 Challenges in Economic Growth Prediction

Predicting economic growth is complex due to several factors:

- Volatility: Economic indicators exhibit significant fluctuations due to market dy-namics and external events.
- Data Quality and Availability: Limited or inconsistent historical data can hindermodel performance.
- Non-Stationarity: Economic time series often exhibit trends and seasonality, com-plicating predictions.
- External Factors: Geopolitical events, pandemics, and policy changes can unpre-dictably affect growth.

This study addresses these challenges by using robust preprocessing techniques and a powerful machine learning model.

5 System Configuration Analysis

5.1 Software Requirements

The project was implemented using Python 3.8 with the following libraries:

- pandas for data manipulation.
- scikit-learn for preprocessing and model evaluation.
- xgboost for building the predictive model.
- shap for feature importance analysis.
- matplotlib and seaborn for visualization.

5.2 Hardware Configuration

The analysis was conducted on a standard laptop with 16GB RAM and an Intel i7 processor, sufficient for training the XGBoost model and generating visualizations.

6 Experimental Analysis & Results

6.1 Data Collection

The dataset, sourced from a CSV file, includes economic indicators (GDP, inflation rate, unemployment rate, and economic growth) for 19 countries from 2010 to 2025. The data was likely compiled from reputable sources like the World Bank or IMF.

6.2 Data Loading and Preparation

The data was loaded using pandas, and preprocessing steps included:

- Checking for missing values and duplicates.
- Encoding the categorical "Country" variable using LabelEncoder.
- Creating lagged features (past 3 years) for GDP, inflation, and unemployment tocapture temporal dependencies.
- Splitting the data into 80% training and 20% testing sets.

6.3 Data Visualization

Visualizations were generated to explore the data:

- Histograms: Showed distributions of GDP, inflation, and unemployment.
- Count Plot: Illustrated the number of records per country.
- Box and Violin Plots: Compared GDP and inflation distributions across countries.
- Correlation Heatmap: Highlighted relationships between variables.

6.4 Model Development for Growth Prediction

The XGBoost regressor was selected for its ability to handle structured data and capture non-linear relationships. Key hyperparameters included:

- objective='reg:squarederror'
- n_estimators=200
- learning_rate=0.05
- max_depth=6

Features included encoded country names and lagged economic indicators. The model was trained on the training set and evaluated on the test set.

6.5 Results

The model achieved the following performance metrics: Visualizations of actual vs predicted economic growth showed reasonable alignment, indicating the model's effectiveness. SHAP analysis revealed that lagged GDP values were the most influential features.

Table 1: Model Performance Metrics

Metric	Value
Mean Absolute Error (MAE)	2.086
Mean Squared Error (MSE)	16.870
Root Mean Squared Error (RMSE)	4.107

7 Future Scope

Future improvements could include:

- Exploring deep learning models like LSTM for time series forecasting.
- Incorporating additional features, such as sentiment indices or geopolitical risk data.• Extending the model to predict other economic indicators or cover more countries.
- Developing real-time forecasting tools for policymakers.

8 Conclusion

This study demonstrates the effectiveness of the XGBoost algorithm in predicting economic growth rates using historical economic data. The model's performance, with an RMSE of 4.107, suggests it can provide valuable insights for economic forecasting. While challenges like data quality and external factors remain, the use of machine learning offers a promising approach to improving prediction accuracy. Future work could enhance the model by incorporating additional data sources and advanced algorithms.

9 References References

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